An Explainable Distributed Deep Learning Framework for Financial Risk Management

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PROBLEM DEFINITION

This project aims to design an Explainable Distributed Deep Learning Framework for Financial Risk Management. The proposed system will analyse different sources of financial information in order to identify patterns related to financial risk, market behaviour, and decision-making in the stock market.

Input:

- Public stock market trading data, including historical prices, trading volumes, volatility indicators, and other time-series market signals.
- Financial news and company reports contain textual information about company performance, market sentiment, economic developments, and investor expectations.
- Tabular financial data, such as company financial indicators, balance sheet variables, revenue statistics, and other structured financial attributes.

These inputs represent different perspectives on financial markets. Time-series data capture the temporal behaviour of market prices, textual news reflects market sentiment and external events, and tabular financial indicators describe company fundamentals.

Output:

The output of the system is a financial risk prediction and analysis. Depending on the task, this output may include:

- A classification label, such as positive or negative market sentiment, or different levels of financial risk.
- A numeric prediction, such as a risk score or probability of a financial event.

In addition to the prediction itself, the system will generate explanations that indicate which features or signals influenced the prediction. This helps improve transparency and trust in the system's decisions.

Learning Type:

The framework combines supervised and unsupervised learning. In supervised learning, the model learns a mapping between input data and labelled outcomes. The Natural Language Processing (NLP) module will use supervised learning to analyse financial news articles and perform tasks such as sentiment classification. The model will learn from labelled examples of positive, negative, or neutral financial sentiment.

Unsupervised learning will be used in other parts of the system. For example:

- Time-series analysis can be used to detect hidden patterns or structures in stock market behaviour without predefined labels.
- Representation learning for tabular financial data will allow the model to learn compressed feature representations that capture important relationships among financial indicators.

Motivation & Relevance

Finance is a highly complex domain where the decisions of investors, banks, and institutions influence the livelihoods of millions of people. The value of stocks, the stability of financial institutions, and the reliability of economic forecasts can all have significant social and economic consequences. Artificial intelligence has increasingly been used in finance to support tasks such as market prediction, fraud detection, investment analysis, and economic forecasting. AI systems can process large amounts of financial data much faster than humans and can discover patterns that may not be visible using traditional statistical methods.

Despite these advantages, the adoption of AI in financial decision-making has been relatively cautious. One major reason is the lack of accountability and transparency in many AI systems. A frequently cited remark from a 1979 IBM training manual states: *"A computer can never be held accountable; therefore, a computer must never make a management decision."* This statement highlights a long-standing concern about relying on automated systems for important financial decisions.

Explainable Artificial Intelligence (XAI) has emerged as an important solution to this challenge. In financial systems, two key requirements exist simultaneously:

1. Predictive capability, which allows models to learn complex patterns from large and diverse datasets such as text, time series, and relational data.

2. Defensibility and accountability, which ensure that models can justify their decisions and comply with regulatory requirements.

Financial institutions operate under strict regulatory environments. Banking supervisors require models used in financial decision-making to be well-documented, validated, and continuously monitored. In addition, regulations concerning data protection and automated decision-making require that individuals affected by algorithmic decisions have access to explanations. In practice, financial machine learning systems also face several technical challenges, including:

- Non-stationarity, where market conditions change over time.
- Distribution shifts, where training data differ from real-world data.
- Class imbalance, especially in tasks such as fraud detection or anti-money laundering.

Because of these challenges, financial AI systems must not only produce accurate predictions but also generate explanations that can be verified and interpreted by human experts. Systems that provide clear and validated explanations are more likely to satisfy governance requirements and be accepted in real-world financial environments.

Deep learning is particularly suitable for this problem because financial data are high-dimensional and complex. According to Goodfellow's discussion of the "curse of dimensionality", traditional models often struggle when data contain many interacting variables. Deep neural networks can learn hierarchical representations of data, allowing them to extract meaningful patterns from large and complex datasets.

Research Questions:

1. Can a distributed deep learning framework that integrates multiple data sources—such as financial time-series data, financial news text, and tabular financial indicators—improve financial risk prediction compared with traditional single-model approaches?
2. Can supervised NLP models effectively capture financial sentiment from company news and improve prediction performance when combined with other financial signals?
3. Can unsupervised representation learning for time-series and tabular financial data reveal underlying patterns that enhance risk prediction and market analysis?
4. Can the integration of explainable AI techniques provide transparent and verifiable explanations for financial predictions, making the system suitable for real-world financial governance and regulatory requirements?

LITERATURE REVIEW

Deep learning is now widely applied in financial tasks such as risk prediction, **c**redit evaluation, fraud detection, and market forecasting. This shift is largely due to the nature of modern financial data, which is massive, noisy, and often highly nonlinear. In the past, most financial systems depended on traditional statistical techniques like logistic regression, ARIMA, or rule-based approaches. Although these models were relatively easy to interpret, they often struggled to capture complex relationships across transaction histories, market indicators, textual information, and network connections.

Recent research shows a clear transition from these traditional approaches to more advanced models such as LSTMs, Transformers, graph neural networks, and multi-agent systems. However, finance is a high-stakes domain where decisions can have significant consequences, so achieving high accuracy alone is not sufficient. Because of this, current studies increasingly

focus on model transparency, user trust, and regulatory compliance. This direction is particularly relevant to our project. It suggests that the most meaningful contribution is not simply building another accurate predictive model, but developing a framework that brings together distributed analysis, deep learning techniques, and explainable results within a single system.

Review of Baseline Papers

1. TradingAgents - Multi-Agents LLM Financial Trading Framework
   Problem & Data:
   This paper examines financial decision-making in a dynamic trading environment. Rather than relying on a single model, it adopts a distributed multi-agent architecture in which different agents are responsible for tasks such as fundamental analysis, sentiment analysis, technical analysis, debate, trading, and risk management. Historical stock prices, news articles, social media signals, insider trading records, financial statements, and technical indicators gathered from significant technology companies are just a few of the data sources that the system uses.
   Main Idea:
   The system deploys multiple LLM-powered agents, each assigned a distinct role mirroring that of a real trading firm. Four analyst agents independently process different information streams: fundamentals, news, sentiment, and technical indicators and deposit their findings into a shared Decision Enabling Warehouse (DEW), which acts as a dynamic knowledge blackboard. A Bull and a Bear researcher agent then engage in a structured multi-round debate, arguing opposing market positions before a trader agent synthesises their conclusions into a directional decision. A three-member risk management team independently challenges this decision, and a fund manager makes the final approval. Communication between agents combines structured protocol messages with natural language, enabling emergent reasoning that adapts to the complexity of each market scenario.
   Training Setup:
   TradingAgents does not undergo conventional gradient-based training on a labeled financial dataset. Instead, it relies on pre-trained LLMs: GPT-4o-mini and GPT-4o for analytical tasks, and o1-preview for deep reasoning
   Results & Metrics:
   Performance is measured using Cumulative Return (CR), Annualized Return (AR), Sharpe Ratio (SR), and Maximum Drawdown (MDD). On AAPL, the system achieves a cumulative return of 26.62% with a Sharpe Ratio of 8.21 and a Maximum Drawdown of just 0.91%, representing a +24.57% improvement over the Buy and Hold baseline. Consistent outperformance over MACD, KDJ+RSI, ZMR, and SMA is reported across all six stocks.
   Strengths:

The framework stands out for its innovative approach to simulating the operations of a real trading firm through collaboration among LLMs. Its integration of multi-modal data, structured debate process, and role-specific agent design marks a significant shift away from traditional single-model methods

Weaknesses:

The three-month test period is too short to judge performance in different market conditions, and the authors note that the high Sharpe Ratios are partly due to a calm market. Using proprietary LLMs makes the system expensive and hard to reproduce

2. Big Data and Machine Learning Methods in Financial Risk Prediction

Problem & Data:

This paper focuses on financial risk prediction, covering areas like credit risk, market risk, and fraud detection, using large financial datasets that include transactions, financial statements, market prices, social media, and news. It looks at big data pipelines across various financial data types rather than depending on a single benchmark, so the findings are applicable to a wide range of financial prediction tasks.

Main Idea:

The paper presents a structured overview of machine learning and deep learning approaches for financial risk prediction.It includes supervised techniques such as deep neural networks and gradient boosting, unsupervised clustering for anomaly detection, graph neural networks for relational financial modeling, and multimodal architectures that integrate numerical and textual sentiment. The main argument is that integrating multiple data types and models produces more reliable predictions than any single approach, and that explainable AI techniques such as SHAP and LIME are essential to make model decisions actionable and compliant with regulations

Training Setup:

This paper refers to standard gradient-based training using appropriate loss functions; cross-entropy for classification and MSE for regression and discusses regularization techniques to reduce overfitting in high-dimensional financial datasets.

Results & Metrics:

The paper summarises performance across the reviewed models using Accuracy, Precision, Recall, F1-score, and domain-specific credit risk metrics. Logistic regression is used as a baseline, and the review consistently shows that ensemble methods and deep learning architectures outperform it in financial risk prediction.

Strengths:

The paper provides a well organised synthesis of current best practices in financial risk prediction. It is practically helpful because of its emphasis on multimodal data integration and explainable AI's role in regulatory compliance, and it highlights new research directions with its discussion of graph-based and privacy-preserving techniques.

Weaknesses:

Its limitation is that it is a survey rather than a single experimental system, so it does not provide a reimplementable model pipeline on its own.

3. GNNExplainer: Generating Explanations for Graph Neural Networks
   Problem & Data:
   This paper focuses on explaining predictions made by graph neural networks. While it does not directly address financial risk, its approach is highly relevant because many financial systems can be represented as graphs, including borrower-guarantor networks, transaction networks, and contagion structures. The method is tested on four synthetic datasets: BA-Shapes, BA-Community, Tree-Cycles, and Tree-Grid, which contain known structural motifs, and two real-world datasets: MUTAG, a molecular graph classification dataset, and Reddit-Binary, a social interaction thread classification dataset.
   Main Idea:
   GNNs generate predictions by aggregating information from a node's neighbours over multiple layers, creating embeddings that capture both graph structure and node features. GNNExplainer operates post-hoc: given a trained GNN and a particular node prediction, it learns a soft mask over the edges and features in the computation graph of that node, which is the subgraph that the GNN actually uses. The mask is optimised to maximise the mutual information between the masked subgraph and the predicted label, effectively highlighting the minimal subset of structure and features responsible for the prediction
   Training Setup:
   The GNN is trained with gradient descent using cross-entropy loss for classification. The GNNExplainer mask is optimised separately, also with gradient descent, by minimising a conditional entropy objective.
   Results & Metrics:
   Explanation quality is measured with Explanation Accuracy, comparing learned mask weights to ground-truth motif edges. GNNExplainer outperforms the gradient-based baseline (GRAD) and the graph attention baseline (ATT) by 17.1% on average across datasets, reaching up to 43.0% improvement on the hardest Tree-Grid dataset.
   Strengths:
   The method is fully model-agnostic, works for node classification, link prediction, and graph classification, and provides explanations for both graph structure and node features.
   Weaknesses:
   GNNExplainer is computationally expensive for large graphs, lacks evaluation on real financial transaction data, and its mutual information objective can struggle with correlated features in noisy datasets. Newer GNN explanation methods have since improved efficiency and accuracy

Review of Additional Papers

1. **Papers that define the problem and the explainability gap**
   Černevičienė and Kabašinskas (2024) in *Explainable artificial intelligence (XAI) in finance: a systematic literature review,* analyse 138 studies focused on explainable artificial intelligence (XAI) in the finance sector, identifying credit scoring, stock prediction, and fraud detection as the predominant use cases. They indicate that although the financial industry is progressively utilising black-box models, stakeholders require explanations that are clear and verifiable, reinforcing our choice to incorporate explainable outputs with high-performing predictors.

   In *Model-Agnostic Explainable Artificial Intelligence Methods in Finance: A Systematic Review, Recent Developments, Limitations, Challenges and Future Directions*, the review 150 studies on model-agnostic XAI in finance, highlighting methods such as SHAP, LIME, counterfactuals, and partial dependence plots. They support our strategy of treating explainability as a fundamental system feature rather than an add-on by arguing that future systems should offer real-time, regulation-aware explanations.

   A Comprehensive Review on Financial Explainable AI provides a broad overview of FinXAI and highlights the key trade-off between model complexity and interpretability. The paper emphasises that explainability is not just a technical feature but a requirement for trust, governance, and high-risk applications, supporting the human-centred aspects of our proposal.

2. **Papers that show additional modelling ideas in the same domain**
   An early multi-agent stock trading decision support system is presented by Luo and colleagues (2002) in their paper: *A Multi-Agent Decision Support System for Stock Trading,* highlighting the long-standing benefits of distributed financial decision-making. Their work influences our design approach by demonstrating how complex financial tasks can be divided into cooperating specialist modules.

   Fraud Detection using Machine Learning and Deep Learning, benchmark several fraud detection methods on European, Australian, and German datasets. Using metrics like AUC, MCC, and cost of failure, they compare KNN, random forest, SVM, autoencoders, CNNs, RBMs, and DBNs, highlighting the importance of taking class imbalance and business cost into account in addition to accuracy when evaluating financial models.

   Explainable AI (XAI) in Deep Learning-Based Financial Forecasting: Improving Model Transparency and Investor Trust by Austin Kate (2019) examines how XAI can be incorporated into deep learning-based financial forecasting using models like CNN, transformers, and LSTM.The study highlights metrics like RMSE, MAPE, directional

accuracy, and economic indicators such as Sharpe ratio and drawdown, reinforcing the importance of evaluating explanations alongside their practical business value.

3. **Paper that supports practical transparent risk scoring**
   In *Explainable Artificial Intelligence Credit Risk Assessment using Machine Learning,* Shreya and Pathak (2025) suggest a transparent credit risk assessment system that combines SHAP and LIME with XGBoost, LightGBM, and Random Forest. The study is useful because it shows how business-facing reports and applicant-level explanations can be produced for lending decisions, even though the predictive models are ensemble machine learning rather than deep neural networks. Our interface design is heavily influenced by this, emphasising that our framework should generate explanations that are comprehensible to managers, end users, and data scientists alike.

PROPOSED APPROACH

This project proposes to develop an explainable distributed deep learning framework for financial risk prediction that combines multiple types of financial information and produces explainable predictions. The idea is to use a deep neural network system that receives different types of financial information and learns patterns from the data through hidden layers, such as market time-series data, textual financial news and financial indicators. Each component processes its input and learns patterns from the data using deep neural networks. The outputs produced are combined in a central decision module. This module passes the combined information through several hidden layers of a neural network in order to learn complex relationships between the different financial signals and outputs a prediction of the financial risk level.

Instead of relying on a single mode, the proposed system uses a distributed structure where different components analyse different types of data. Financial markets are influenced by many factors such as historical price movements, company financial indicators, and external news events. Therefore, the model will learn from multiple data sources and combine their insights to produce a final risk prediction.

In addition to predictions, the proposed framework incorporates explainable Artificial Intelligence. These methods will be used to identify/ highlight the most important features influencing the predictions. This helps financial analysts to understand the reasoning behind the model's decisions.

Input Representation

Financial information arrives from multiple sources and must be converted into numerical representations that can be processed by the neural network

1. **Financial Time-Series Data**

Stock market data, such as historical prices, trading volume, and volatility indicators, are represented as time-series sequences. These values will be organised into ordered sequences of part observations. By analysing these sequences, the model can learn patterns related to trends,cycles and sudden changes in market behaviour

2. **Tabular Financial Data**
   Tabular data is structured financial indicators, for example, company revenues, profit margins and debt ratios will be represented in a tabular form. These features will then be converted into numerical factors and normalised so that their values can be used to compare. This will allow the model to learn relationships between different financial indicators more effectively.
3. **Textual Financial Data**
   Most of the valuable textual information is gained from news articles, company announcements and market commentary. This textual information will be transformed into numerical data using word indices or word embeddings, which represent words as vectors. This allows the model to capture semantic relationships and perform sentiment analysis on financial news.

Network outline

Multiple models are going to be involved in the proposed solution; the cumulative number of layers across the framework will be large. Each module processes a specific type of financial data and consists of several hidden layers that learn patterns from the input. Non - linear activation functions will be used throughout the network so that the models can capture complex relationships within the financial data gained from the different input layers. Since the system follows a multi-agent structure, different models will operate in parallel and analyse different types of information simultaneously. Although the overall architecture is large, computational efficiency will be maintained through parallel processing across the agents. Each model will be trained using gradient-based optimisation with a loss function.

Why is this suitable?

This is a suitable distributed approach designed to allow each deep learning model to focus on a specific task. Financial data comes from multiple sources, such as time-series market data, financial news and tabular financial indicators, and separate models can analyse each type more effectively. This architecture improves efficiency and prediction performance because each model specialises in learning patterns within its own domain. explainability techniques such as SHAP and LIME can be applied more effectively to smaller individual models rather than a single large model. This provides greater transparency and accountability, allowing users to better understand how the system reaches its predictions.

DATASET DESCRIPTION

Source of data

- **Kaggle**: “Huge Stock Market Dataset” by Boris Marjanovic, “Home Credit Default Risk” by Megan Crenshaw, “Give Me Some Credit :: 2011 Competition Data” by Bryce Freshcorn, “German Credit Dataset” by CaramellDogo, among others.
- **yfinance (Python Library)**: This is a "scraper" for Yahoo Finance.
- **Common Crawl**: Archives for historical web news data.
- **FRED (Federal Reserve Economic Data)**: Python API (fredapi).
- **Polygon.io:** They have high-quality historical "tick" and "day" data.
- **EODHD (End-of-Day Historical Data):** They have some US stocks going back to the 1970s and even earlier for major indices.

Size and Structure

The datasets contain categorical, textual, and numerical features, with each dataset comprising thousands of samples.

Preprocessing

Missing values will be handled by either imputing them or retrieving the information from alternative data sources. Time-series and categorical features will be normalized to ensure consistency, while textual data will undergo minimal preprocessing, such as the removal of stop words.

Why these datasets fit the problem

These datasets cover all domains relevant to our models and align with the phenomena we aim to study. They are also sufficiently large to support deep learning models and ensure reproducibility of results.

EVALUATION PLAN

Metrics

Each module of the framework will be evaluated using AUC-ROC as the primary performance metric, given its robustness to class imbalance in financial datasets, alongside F1-score to reflect the trade-off between precision and recall. For explainability quality, a faithfulness metric will be applied to assess whether the SHAP and LIME explanations genuinely reflect each model's decision process specifically, whether removing the top-ranked features causes a proportionate drop in model confidence. SHAP will provide global feature importance across the training data while LIME will generate local explanations for individual predictions.

Experimental setup

The full available dataset will be used for training and fine-tuning each module of the distributed framework. Testing will be conducted on live data fetched through real-time

APIs, including yfinance, FRED, and Polygon.io, rather than a static held-out split. This reflects the intended deployment context of the system and provides a more demanding evaluation, as live data is subject to distribution shifts and market regime changes that fixed test sets do not capture.

Baseline comparison

Per-model performance metrics will be compared against those reported in the corresponding baseline papers for each subtask. At the system level, one direct comparison will be made against Baseline 1 : TradingAgents (Xiao et al., 2025) , using shared metrics including cumulative return, Sharpe ratio, and maximum drawdown, to assess whether the proposed framework achieves competitive performance against the state-of-the-art multi-agent approach.

EXPECTED OUTCOMES

The proposed framework targets measurable improvements in financial risk prediction by drawing on complementary data sources such as time-series market data, financial news, and structured financial indicators. Integrating these inputs into a unified architecture addresses a fundamental shortcoming of single-modality models, which are mostly constrained by the limited information available from any one data type. By operating across multiple modalities simultaneously, the framework is designed to produce richer feature representations and, consequently, stronger predictive performance. An advantage of a distributed design is task specialisation. Rather than requiring a single model to process heterogeneous data, each component focuses on a distinct information stream. This division is intended to improve the quality of learned representations for each data type, yielding combined outputs that outperform any individual component in isolation. The integration of explainability techniques serves a secondary but important objective: making the model's decision-making process interpretable to analysts

Learning Outcomes

The project allows us to develop an understanding of several core deep learning principles. Working with multi-layer architectures will provide us with direct insight into hierarchical representation learning, specifically, how successive layers encode increasingly abstract features of financial data, capturing non-linear dependencies that conventional models cannot represent. This clarifies the relationship between model depth and predictive capability. The project will also show the advantages of learned feature extraction over hand-engineered indicators. Where traditional approaches rely on predefined signals, deep models discover patterns directly from raw data, potentially revealing structure that domain expertise alone would overlook. Observing this contrast

in a financial context will sharpen understanding of when and why learned representations add value. Finally, systematic experimentation with hyperparameters, learning rate, layer count, and hidden dimension size will build intuition for model optimisation and training stability, areas that are difficult to appreciate without hands-on experience.

Limitations

Firstly, the quality and availability of publicly available financial data is uneven, real markets involve dynamics that no public dataset fully captures and limited data volume may reduce the model's ability to generalise to novel market conditions.

Secondly, operating multiple models within a distributed architecture places substantial demands on computational resources. This may necessitate reducing the scale of experiments, for instance, by restricting the range of hyperparameter searches or the size of training datasets.

Thirdly, time and resource constraints will require that some aspects of the baseline methods or multi-agent frameworks be approximated rather than replicated precisely. Where full implementation is not feasible, the project will document these simplifications and account for their potential effect on comparative results. These constraints are acknowledged upfront to ensure that conclusions are drawn within an appropriate scope. Despite them, the project offers a substantive contribution to understanding how distributed deep learning architectures can be applied to financial risk analysis, and how explainability tools can support transparency without sacrificing model utility.

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